WEEK 1

Task1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Source Code:

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main(){
    char command[100];
    pid_t pid;
    printf("Enter a Linux command: ");
    scanf("%s", command);
    pid = fork();
    if(pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if(pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());

        execlp(command, command, NULL);
        printf("Command execution failed.\n");
        exit(1);
    } else {
        wait(NULL);
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child process completed.\n");
    }
    return 0;
}
```

Output:

```
Enter a Linux command: ls

Child Process
Child PID : 437
OSSP  code.c  files.c  gm.c   hello.c    os    program.c    programming.c       source_code
code  files   gm       hello  operating  os.c  programming  programming.c.save  source_code.c

Parent Process
Parent PID : 436
Child process completed.
```

Task2

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Report

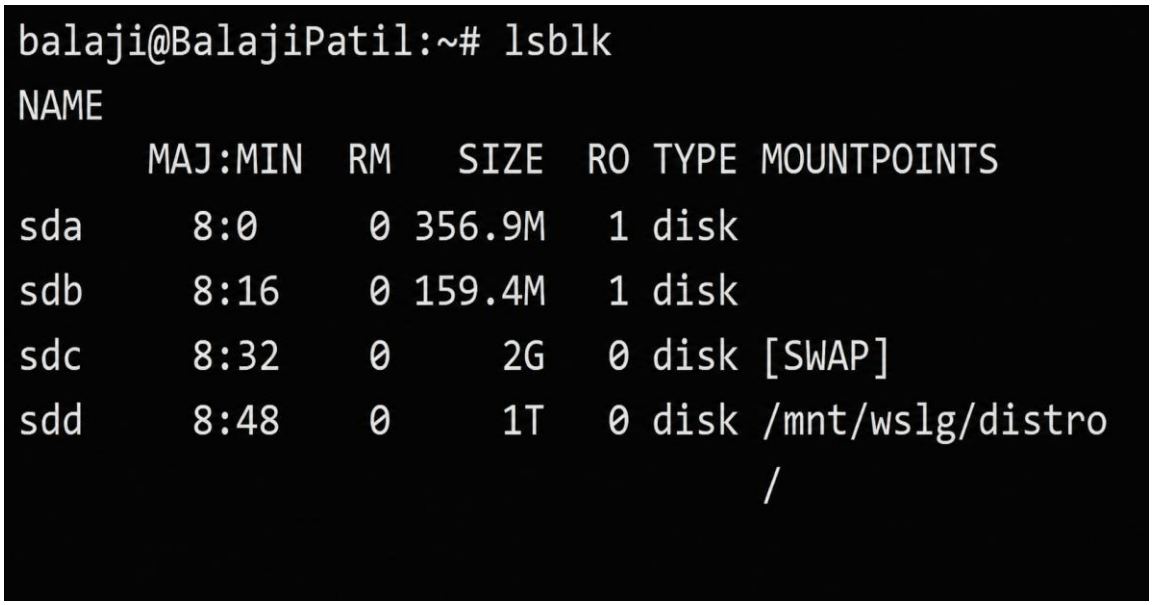
1. Uname: Displays operating system and kernel information.

```
balaji@BalajiPatil:~# uname
Linux
```

2. lscpu: Displays CPU information.

```
balaji@BalajiPatil:~# lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:             39 bits physical, 48 bits virtual
  Byte Order:                Little Endian
CPU(s):                      4
  On-line CPU(s) list:       0-3
Vendor ID:                   GenuineIntel
  Model name:                Intel(R) Core(TM) i3-8145U CPU @ 2.10GHz
    CPU family:              6
    Model:                   142
    Thread(s) per core:      2
    Core(s) per socket:      2
    Socket(s):               1
    Stepping:                12
    BogoMIPS:                4608.01
```

3. lsblk: Displays storage devices.



4.ps: Displays currently running processes.

```
balaji@BalajiPatil:~# pa
    PID TTY          TIME CMD
    352 pts/1    00:00:16 bash
    946 pts/0    00:00:00 pa
```

4.top: Displays real-time system performance.

```
balaji@BalajiPatil:~# top
top - 13:56:44 up 22 min,  1 user,  load average: 0.06, 0.03, 0.00
Tasks:  22 total,   1 running,  21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7880.6 total,   7463.6 free,    438.7 used,    123.2 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7441.9 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root      20   0   21680  13140   9744 S   0.0   0.2   0:00.74 systemd
    2 root      20   0    3180   2200   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    9 root      20   0    3196   2128   2008 S   0.0   0.0   0:00.00 init
   64 root      19  -1   34052  12556  11488 S   0.0   0.2   0:00.25 systemd-journal
  112 root      20   0   25548   6924   5168 S   0.0   0.1   0:00.26 systemd-udevd
  123 systemd+  20   0   21344  13252  11124 S   0.0   0.2   0:00.12 systemd-resolve
  124 systemd+  20   0   91036   7992   7024 S   0.0   0.1   0:00.10 systemd-timesyn
  168 root      20   0    4244   2688   2432 S   0.0   0.0   0:00.00 cron
  169 message+  20   0    9540   5432   4736 S   0.0   0.1   0:00.09 dbus-daemon
  176 root      20   0   17980   8888   7848 S   0.0   0.1   0:00.08 systemd-logind
  178 root      20   0 1756104  13024  10888 S   0.0   0.2   0:00.14 wsl-pro-service
```